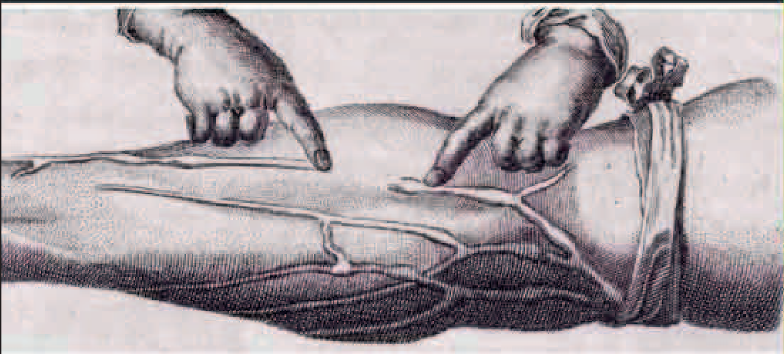
1631–34



William Harvey demonstrated that one-way valves in veins stop blood from flowing back to the hand.

Galileo Galilei was tried by the Inquisition and forced to retract his heliocentric views. He was placed under house arrest, where he remained till he died.

WILLIAM HARVEY (1578–1657)

Born in England, William Harvey studied at the universities of Cambridge and then Padua, in Italy, before a career in England devoted to studying blood and circulation; later, he also investigated reproduction and development. He served as physician to both King James I and King Charles I, and treated victims of the English Civil War.

a collection of machine designs that included an **early steam engine**. The steam-blasting vessel blew through a pipe that was directed at the vanes of a paddle-wheel, causing the wheel to revolve. Branca came up with several uses for his machine: lifting water, and grinding stone or gunpowder. In reality, however, it would have limited practical use. It was also entirely unrelated to later, more successful steam engine designs.

Englishman **John Parkinson** (1567–1650) was a herbalist and apothecary to the king. He was also a plantsman caught between the ancient herbalists and a new generation of botanists. His first major horticultural book—wryly entitled *Park-in-Sun's Terrestrial Paradise*—became an important text on the cultivation of plants, both for their esthetic and medicinal qualities. Although widely acknowledged as the **first gardening book**, it made limited impact on the scientific understanding of plants.

800
THE APPROXIMATE NUMBER OF **PLANTS ILLUSTRATED** IN PARKINSON'S 1629 WORK

“AND YET IT **MOVES.**”

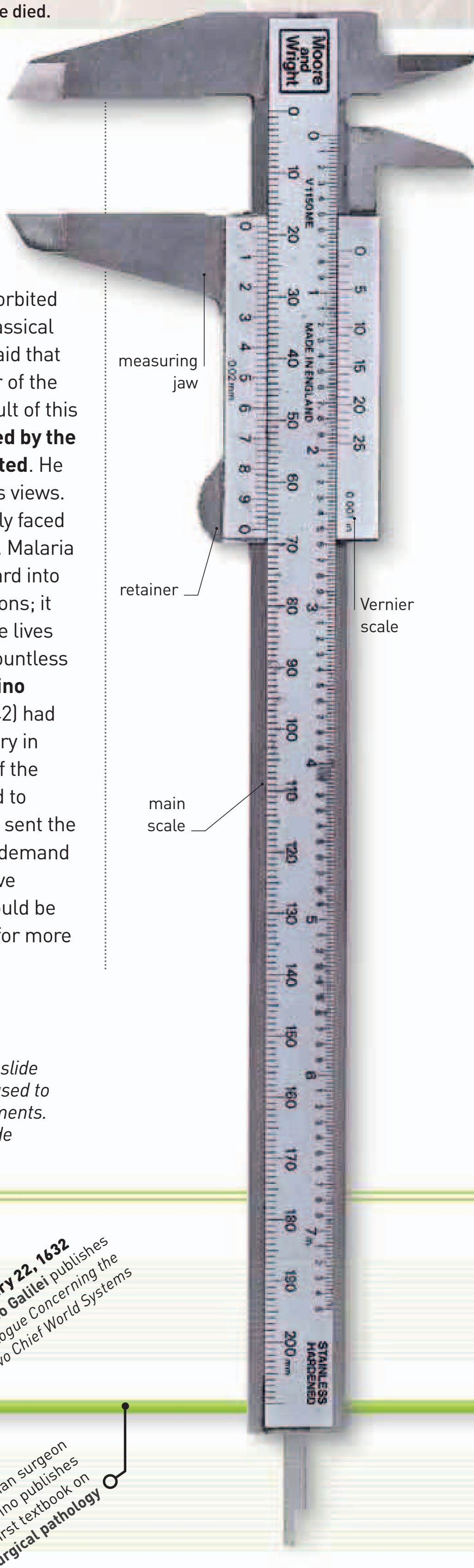
Galileo Galilei, Italian astronomer, supposedly after his forced recantation of the theory that Earth moves around the Sun, 1633

IN 1631, FRENCH MATHEMATICIAN PIERRE VERNIER (1580–1637) described a device that assisted in the accurate **measurement of length**. It was based on an earlier idea by German mathematician Christopher Clavius. The original instrument had a sliding scale along the edge of a quadrant, which meant the user could measure a fractional part of the smallest division on the scale. To this day, the **Vernier scale** remains one of the best mechanical devices for accurate measurement.

In the same year, English mathematician **William Oughtred**—inventor of the slide rule—published a text that would have a lasting influence on many other mathematicians, including Isaac Newton. Oughtred's ***The Key of the Mathematicks*** introduced some fundamental algebraic symbols: the multiplication sign (x), and the proportion sign (:). For years it was described as the **most influential mathematical publication** in England.

Early in 1632, Italian astronomer **Galileo Galilei** published *Dialogue Concerning Two Chief World Systems*. In it he defended the heliocentric model of Copernicus that Earth orbited the Sun, against the classical view of Ptolemy, who said that Earth was at the center of the Solar System. As a result of this heresy, Galileo was **tried by the Inquisition and convicted**. He was forced to recant his views.

In the early 1630s, Italy faced a deadly natural threat. Malaria was spreading northward into swampy, low-lying regions; it had already claimed the lives of several popes and countless Roman citizens. **Agostino Salumbrino** (1561–1642) had worked as an apothecary in Peru, where the bark of the cinchona tree was used to control the disease. He sent the bark to Europe, where demand for it escalated. Its active ingredient, **quinine**, would be used to **treat malaria** for more than 300 years.



Vernier scale
Two adjoining scales that slide against one another are used to make accurate measurements. This device helps subdivide the smallest of divisions.

1629 Neil O'Glacan publishes a **treatise on plague**, after tending its victims

1631 Pierre Vernier first describes a **Vernier scale**

1631 English mathematician William Oughtred introduces **algebraic symbols**

1631 Peruvian Agostino Salumbrino sends **quinine-containing bark** to Rome to treat **malaria**

February 22, 1632 **Galileo Galilei** publishes *Dialogue Concerning the Two Chief World Systems*

1632 Italian surgeon Marco Severino publishes first textbook on **surgical pathology**